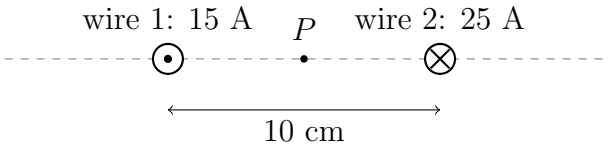
Test 4 Practice — Magnetism & Induction

These are modeled on the hardest problems from Written HW 7, HW 8, and HW 9 – same problem types, new numbers. Every formula you need is on the Magnetism equation sheet; have it open while you work. Each pair of problems below tests the same idea but asks for a different unknown, so skim the whole question before you start plugging in. The answer key is on the last page.

Part A — Magnetic Fields, Forces & Motion

1. Two long, straight, parallel wires are 10 cm apart. Wire 1, on the left, carries 15 A out of the page; wire 2, on the right, carries 25 A into the page. (a) Find the magnitude and direction of the magnetic field at point P, midway between the wires. (b) At what point on the line through the two wires is the magnetic field zero?



2. Two long, straight, parallel wires are 7.0 cm apart and carry currents in the *same* direction. Wire 1 carries 8.0 A. The magnetic field is zero at a point between the wires, 2.0 cm from wire 1. (a) What current flows in wire 2? (b) What is the magnetic field strength at the point midway between the wires?

3. A 50-turn circular coil of radius 4.0 cm carries a current of 2.0 A. Find (a) the coil's magnetic dipole moment, (b) the magnetic field strength at the center of the coil, and (c) the largest torque the coil can feel when it is placed in a uniform 0.30 T magnetic field.

4. A single-turn circular loop of radius 3.0 cm carries current I. The magnetic field at the loop's center is measured to be 42 μT. (a) What current flows in the loop? (b) What is the loop's magnetic dipole moment? (c) The loop is now placed in a uniform 0.80 T external field. In which orientation is its potential energy greatest, and what is the potential energy in that orientation?

5. A rectangular 20-turn coil, 5.0 cm $\times$ 8.0 cm, carries a current of 1.5 A. It sits in a uniform 0.40 T magnetic field, and the *plane* of the coil makes a 60° angle with the field. Find (a) the magnitude of the torque on the coil, and (b) the coil's potential energy. (c) The coil is released and turns until its dipole moment lines up with the field. By how much does its potential energy change?

6. A small bar magnet feels a torque of 0.036 N·m when its axis is at 35° to a uniform 0.15 T magnetic field. (a) What is the magnitude of the magnet's dipole moment? (b) How much energy does it take to turn the magnet from pointing along the field to pointing opposite the field? (c) At what angle from the field is the magnet's potential energy +0.025 J?

7. An electron travels at 2.0×10^6 m/s between two parallel plates separated by 2.0 cm and charged by a 400 V battery (top plate positive). What magnetic field strength and direction will let the electron pass between the plates without being deflected?

8. An antiproton (same mass as a proton, $q = -e$) moves at 400 m/s to the left through a region with uniform fields $\vec{E} = 1500$ V/m (pointing up, in the plane of the page) and $\vec{B} = 3.0$ T (into the page). Find the magnitude and direction of the antiproton's acceleration at this instant.

9. A uniform 40 mT magnetic field points in the $+z$ direction. A proton enters the field moving at 2.0×10^6 m/s, at 40° above the xy-plane. Find the radius r and the pitch p of the proton's helical trajectory.

10. A proton moves through a uniform 25 mT magnetic field on a helical path with radius 2.0 cm and pitch 8.0 cm. Find the proton's speed v and the angle above the xy-plane at which it entered the field.

Part B — Electromagnetic Induction

11. A 15-cm-long wire is pulled along a U-shaped conducting rail in a perpendicular magnetic field. The total resistance of the wire and rail is 0.40 Ω. Pulling the wire at a steady speed of 3.0 m/s causes 6.0 W of power to be dissipated in the circuit. Find (a) the pulling force, and (b) the magnetic field strength.

12. A 20-cm-long wire slides along a U-shaped conducting rail in a 1.5 T field perpendicular to the rail's plane, pulled at a constant 5.0 m/s. The total resistance is 0.60 Ω. Find (a) the induced emf and current, (b) the power dissipated, and (c) the pulling force required.

13. A square conducting loop, 15 cm on each side, is positioned so that exactly 60% of its area lies within a 0.25 T magnetic field (directed into the page); the rest of the loop is in a field-free region. (a) What is the magnitude of the magnetic flux through the loop? (b) If the field strength *increases*, what is the direction of the induced current in the loop?

14. A bar magnet's south pole is pushed toward a stationary conducting loop, face-on. As the magnet approaches, is a current induced in the loop? If so, use Lenz's law to determine whether the near face of the loop becomes a north or south magnetic pole.

15. A 500-turn coil of wire 2.0 cm in diameter sits in a magnetic field that increases from 0.050 T to 0.200 T in 25 ms. The coil's axis is parallel to the field. What is the induced emf of the coil?

16. A 50-turn coil, 4.0 cm in diameter, has a resistance of 2.0 Ω. A magnetic field perpendicular to the coil increases at a steady rate of 0.80 T/s. Find (a) the induced emf, and (b) the induced current.

17. The magnetic field inside a circular region of radius 4.0 cm points into the page and is decreasing at the rate 0.25 T/s. (a) What is the magnitude of the induced electric field at a point 2.0 cm from the center? (b) Is the induced electric field clockwise or counterclockwise? (c) A proton released from rest at that point momentarily feels a force. What is the magnitude of its initial acceleration?

18. At a point 5.0 cm from the center of a circular field region, the induced electric field has magnitude 0.030 V/m. (a) At what rate is the magnetic field changing? (b) An electron released from rest at a point 2.0 cm from the center (same field region) feels what initial acceleration?

ANSWER KEY — FLIP THE PAGE OVER (OR STAND ON YOUR HEAD)

Part A — Magnetic Fields, Forces & Motion

1. (a) $B = B_1 + B_2 = 60\ \mu\text{T} + 100\ \mu\text{T} = 160\ \mu\text{T}$, toward the top of the page (b) 15 cm to the left of wire 1 (25 cm from wire 2)

2. (a) $I_2 = 20$ A (b) $B = 114\ \mu\text{T} - 46\ \mu\text{T} \approx 69\ \mu\text{T}$

3. (a) $m = NIA = 0.50\ \text{A}\cdot\text{m}^2$ (b) $B = 1.6$ mT (c) $\tau_{\max} = mB = 0.15$ N·m

4. (a) $I = 2.0$ A (b) $m = 5.7\times10^{-3}\ \text{A}\cdot\text{m}^2$ (c) $\vec{m}$ opposite to $\vec{B}$; $U = +mB = 4.5$ mJ

5. $m = NIA = 0.12\ \text{A}\cdot\text{m}^2$, and $\vec{m}$ is 30° from $\vec{B}$ (it is perpendicular to the coil's plane) (a) $\tau = 0.024$ N·m (b) $U = -0.042$ J (c) it drops by 6.4 mJ ($\Delta U = -6.4\times10^{-3}$ J)

6. (a) $m = 0.42\ \text{A}\cdot\text{m}^2$ (b) $\Delta U = 2mB = 0.13$ J (c) $\theta \approx 113°$

7. $B = 10.0$ mT, into the page

8. $F_{\text{net}} = 4.8\times10^{-17}$ N, down; $a = 2.9\times10^{10}\ \text{m/s}^2$, down

9. $r = 0.40$ m; $p = 2.1$ m

10. $v \approx 5.7\times10^{4}$ m/s; $\theta \approx 33°$

Part B — Electromagnetic Induction

11. (a) $F_{\text{pull}} = 2.0$ N (b) $B \approx 3.4$ T

12. (a) $\mathcal{E} = 1.5$ V, $I = 2.5$ A (b) $P = 3.75$ W (c) $F_{\text{pull}} = 0.75$ N

13. (a) $\Phi \approx 3.4\times10^{-3}$ Wb (b) counterclockwise

14. Yes; the near face becomes a south pole (repelling the approaching magnet)

15. $\mathcal{E} \approx 0.94$ V

16. (a) $\mathcal{E} \approx 50$ mV (b) $I = 25$ mA

17. (a) $E = 2.5\times10^{-3}$ V/m (b) clockwise (c) $a \approx 2.4\times10^{5}\ \text{m/s}^2$

18. (a) $|dB/dt| = 1.2$ T/s (b) $a \approx 2.1\times10^{9}\ \text{m/s}^2$